

Essentialism and the Essence of to be Necessary

In his 1994 paper *Essence and Modality* Kit Fine argued against the view that essence should be understood in terms of *de re* modality. He suggested instead to understand metaphysical necessity by essence of things. The idea to explain, especially to ground metaphysical necessity in essence was followed by various philosophers albeit with some differences in details. I term all such accounts *essentialism* on necessity.

To ground necessity on essence, essentialists have to provide an account on essence that does not rely on modal notions. And at the same time, they must also derive the necessity of p from the essentiality of p . Some philosophers argue that this has never been successfully established. The arguments provided for the latter implicitly assume some modal characteristic on the essentiality of p , which is to fail to satisfy the former requirement. And if the essentiality of p has no modal import at all, there seems to be a gap between what it is to be essential and what it is to be necessary.

In this paper, I will argue that essentialists can reply to the challenge of gap without compromising their core theses. One remaining possibility is overlooked: the alleged gap is closed by the essence of *to be necessary*. More specifically, I argue that both requirements can be satisfied by assuming that it is essential of *to be necessary* that if something is essential of some objects, then it is necessarily so.

In section 1, I will introduce the problem of the source of necessity, and argue all essentialist accounts give non-transmissive explanation to necessity, even if it was not explicitly stated. The challenge of gap will be elaborated in detail in section 2. I argue in section 3 that what the challenge actually shows is that any account on the source of necessity that utilizes non-transmissive explanation must involve certain commitment to the essence of *to be necessary*. It would be incorrect to conclude from the challenge that essentialism is untenable. In section 4, I will clarify why my suggestion is legitimate. Especially, I argue that

it does not face the problem of circularity. Note that I am not trying to argue for the essentialist answer for the problem of the source of necessity, against other alternatives.

1 The Source of Necessity and Non-Transmissive Explanations

For any necessary truth p , we can ask *why* is p necessary. It is different from the question of why p is true. For example, we can answer the question of why is $(3^2 = 9 \vee \text{snow is white})$ true by saying that $3^2 = 9$ is true. But this does not explain why the disjunction is necessary.

Similarly, the question of the source of a *particular* necessity should be distinguished from the question of the *general* source of necessity. The particular necessity of the previous disjunction might be explained by the necessity of $3^2 = 9$. This however does not guide us at all about the general source of necessity, since the necessity of $3^2 = 9$ is left unexplained.

Blackburn (1993) argued that the general source of necessity cannot be given by some true propositions. Suppose we want to claim that $\Box p$ because q . If q is necessary it cannot be the final source of necessity, for it leaves the necessity of q unexplained. But q also cannot be contingent, for “... then there is strong pressure to feel that the original necessity has not been explained or identified, so much as undermined.” (Blackburn 1993, p.53)

Blackburn wishes to conclude that therefore any realist conception on necessity fails. This advance might be too hasty (Cameron, 2010). However, it was also argued that it is indeed possible to explain necessity via contingent or necessary truths. Hale (2002) pointed out that denying the contingency horn presupposes that necessity is never contingent, which some philosophers, notably conventionalists would disagree. But Hale himself took the other option. He argued that we can explain necessity by necessary truths, insofar as the explanation is *non-transmissive*. A transmissive explanation takes the form of $\Box p$ because $\Box q$, where q itself cannot explain $\Box p$. Perhaps Blackburn is correct if this is the only available form of explanation via necessary truths. A general scheme for giving transmissive explanations may help us to sort out so-called “fundamental” necessities from non-fundamental ones, but does not offer a theory of general source of necessity.

Non-transmissive explanations, by contrast, take the form of $\Box p$ because q , where q is necessarily true “but in which what explains the necessity of the

explanandum is not q 's *necessity*, but its *truth simpliciter*." (Hale 2002, p.311) Here the regress does not occur. Since q is also a necessary truth, the necessity of $\Box q$ would also be explained via the truth of a proposition which is necessarily true, say r . But that does not imply that the necessity of p is thus explained by r . For, it is the truth *simpliciter* of q that explains the necessity of p , not the necessity of q .

Therefore, a general scheme of giving non-transmissive explanation to all, or at least every fundamental necessities suffices as an account for the general source of necessity. This is typically done by specifying what condition must an arbitrary p satisfy to be necessary. Hale's preferred view is that the necessity of any proposition p is explained by it being essential to some objects $X_1, \dots, X_n$ (Hale 2013, p.152).¹ There are other candidates of source. For instance, possible world realists would assert that $\Box p$ because p is true in all possible worlds. While it would be necessary that p is true in all possible worlds, it is not the necessity doing the explanatory work. I will classify all those theories of the source of necessity which utilizes non-transmissive explanation as *non-transmissivism*.

Before proceeding, let me clarify myself between various senses and kinds of essence. By "essence" I mean *mediate and consequential* essence, following Fine (1995a). Suppose we have a clear idea of a strict and direct essence, which is immediate and constitutive. Then, mediate essence of F includes both the nature of itself and the nature of things that are involved in the nature of F ; consequential essence of F includes what logically follows from the nature of F .

Also crucial is the distinction between objectual essence and generic essence. On the one side there is the essence of a particular object, and the other side the essence of *to be* such and such. Fine (2015) argued the essence of *to be* F is different from the essence of the concept F . Suppose the objectual essence of the concept *concept* includes the fact that *concept* is a concept. Then it is a consequential essence of it that $\exists x(x \text{ is a concept})$. However, it seems quite implausible to think that it is by nature of *what it is to be a concept* that there is something satisfies it. That is, while $\exists x(x \text{ is a concept})$ is plausibly an essence of the concept *concept*, it would not be an essence of *to be a concept*. This, we may say, illustrates the difference between the objectual essence of concept and generic essence of *to be a concept*.

It would be clear that the essence of objects or things would be the objectual

¹I use two terms "true in virtue of the nature of", "be an essence of" and "essential to" interchangeably, following Fine and Hale. I sometimes use "essence" to denote essential facts or truths, whereas "nature" is used strictly for the non-further-entity sense of essence (Lowe 2008).

essence of that object. And following Fine (1995a, 2015), by $E_F p^2$ I mean p is a generic essence of F . Objectual essence will be notated by using the *rigid predicate* $\lambda x.x = a$, where “ a ” is a name for an object. To say p is an objectual essence of a I write “ $E_{\lambda x.x=a} p$ ”. I will occasionally write E_a to denote the objectual essence of a , but this is merely an abbreviation.

Note that there can be slightly different views among essentialists on the relation between essence and necessity. For Kit Fine, to be metaphysically necessary is *defined* as to be essential to *all objects* (Fine 1994, 1995a). Fine (2015) makes it explicit that such a definition which gives the constitutive necessary and sufficient condition captures both ground and essence. Therefore, although not explicitly stated, Fine would agree that the necessity of p is grounded on p ’s being essential to all objects. By contrast, Hale explicitly claims his account to be non-reductive. Although he agrees with Fine that metaphysical necessity is grounded in the nature of things, he also takes modality to be “fundamental and irreducible” (Hale 2013, Ch.3).

However, regardless of the difference, any sensible essentialist account should also be non-transmissivist. Since essentialists hold that the necessity of p is explainable by it being *objectually* essential to something,³ they are automatically committed to the position that the source of necessity is also necessary. Under the standard axioms of the logic of essence, it is a theorem that $E_F p \rightarrow E_F E_F p$, where F is rigid (Fine 1995a, p.255, see THEOREM 6 (ii)). That is, if p is essential to some object, the fact that p is essential to the object is essential to the object in question. Since essentialists should accept that p is necessary if p is essential to some objects (this will be argued shortly), $E_F p$ which is the source of $\Box p$ must also be necessary. Also remind that essentialists suggest $E_F p$, not the necessity of $E_F p$ to be the explanans of $\Box p$. Thus, essentialists turn out to be holding that the source of necessity is necessarily true yet what does the explanatory work is its *truth* simpliciter, not its necessity. And that is exactly what non-transmissivism is.

2 The Challenge of Gap

One constraint for all essentialist accounts is that *to be objectually essential* should be more fundamental, or metaphysically prior than *to be necessary*. For

²I use “ E ” as the essentialist operator, substituting Fine (1995a)’s notation $\Box$. “ $\Box$ ” is exclusively used as a necessity operator, or a constant one-place higher-order predicate “... is necessary”.

³Fine (1994): “The metaphysically necessary truths can then be identified with the propositions which are true in virtue of the nature of all *objects* whatever” (p.9, italics added); Hale (2013): “Metaphysical necessities might plausibly be identified as those which hold in virtue of the nature of *things* in general.” (p.150, italics added).

example, suppose for Fa to be the essence of a just is for a to necessarily have F . Then it is unsatisfactory to say the necessity of Fa is explained by Fa being essentially true. It just amounts to say Fa is necessary because a necessary has F , and that leaves the *de re* necessity unexplained. It is parallel to the reason why possible world realists should not define to be a possible world as to be a world that is metaphysically possible. But then, how can essentialists guarantee that all essential facts are necessary?

All essentialists must accept that all objectually essential truths are necessary. To acknowledge contingent objectual essences, one must give a criterion demarcating them from necessary objectual essences without appealing to the notion of essence. That is, one must identify the non-essentialist source of necessity, which contingent essences lack. And this apparently undermines essentialism.

Given such a constraint, some philosophers question how can essentialists guarantee all essential truths are necessary. To put it formally, essentialists are committed to what Mackie (2020) calls “Necessity Principle (NP)”:

$$(NP) \quad \forall F \forall p (E_F p \rightarrow \Box p)$$

where F is rigid.⁴ I will occasionally omit one or two universal quantifiers for convenience and readability.

One common understanding of essence among essentialists is by *real definition*: roughly, the constitutive and immediate essence of certain thing or *to be* such and such is captured by their real definition. (Fine 1994, p.2; Hale 2013, p.132; Lowe 2012, p.919). Alternatively, Correia and Skiles (2017) suggested that essence should be understood in terms of generalized identity, such as “for a thing to be a bachelor is for it to be an unmarried adult male” or “for the Atlantic Ocean to be filled with water is for it to be filled with H_2O molecules” (p.643). Here to be F expresses the full essence of to be G if and only if to be G is to be F ; it expresses partial essence of to be G if and only if there is some H such that to be G is to be both F and H (pp.649-50)⁵. The charge is that there seems to be a gap between real definition or generalized identity and metaphysical necessity, therefore (NP) fails. Of course, essentialists may contend that (NP) is obviously true, although is arguable whether such an intuition is robust enough. However, the problem here is not epistemic but metaphysical.

⁴This formalization follows Fine (2020).

⁵Here again the difference between objectual and generic essence will be captured by the difference between rigid predicates and non-rigid predicates. Suppose the *Con* is the predicate *to be a concept*. And using $\equiv$ as the generalized identity operator, $Con(x) \equiv_x p$ states that p is the full generic essence of to be a concept; $[\lambda x.x = Con]y \equiv_y q$ states that q is the full objectual essence of the concept *concept*.

It is unclear how non-modal conception of essence, which essentialists should stick to, can give metaphysical explanation of why (NP) holds. That is, how can essentialists “deliver a modal rabbit out of a non-modal hat” (Mackie 2020, p.252)?

Hale tried to give an argument. Suppose $\Phi\alpha$ is an objectual essence of α . Now suppose it is possible that $\neg\Phi\alpha$. It implies that there is a β such that $\alpha = \beta \wedge \neg\Box\Phi\beta$. But β cannot be identical to α in the world such that $\neg\Phi\beta$. For, it does not satisfy the condition to be identical to α , which is Φ . Therefore by the necessity of identity $\neg\alpha = \beta$, contradiction. Since contradictions are impossible, we can conclude that it is not possible that $\neg\Phi\alpha$. In other words, if $E_{\lambda x.x=\alpha}\Phi\alpha$, then $\Box\Phi\alpha$ (Hale 2013, p.133).

But it has been pointed out that the argument does not successfully address the challenge (Romero 2019, pp.125-7; Leech 2021, pp.893-5). For the proposed argument to succeed, it must be assumed that Φ is *necessarily* the condition for to be identical to α . Either $\Box E_\alpha\Phi$ or $\Box\Phi\alpha$ is required. But what is assumed is only that Φ is *actually* the condition of the identity. To hold that the assumption implies $\Box\Phi\alpha$ simply begs the question. And while $E_\vee(E_{\lambda x.x=\alpha}\Phi)$ ⁶ is provable from $E_{\lambda x.x=\alpha}\Phi$, the inference by Finean definition from $E_\vee p$ to $\Box p$ is problematic, as we will see below. Moreover, Hale himself would not agree to the definition.

In fact, one can construct a logical proof for (NP) once she accepts the Finean definition of necessity. It is an axiom in the logic of essence that $F \subseteq G \rightarrow (E_F p \rightarrow E_G p)$, where $R \subseteq G$ is for $\Pi x(Rx \rightarrow Gx)$ (Fine 1995a, pp.246-7), and Π is a quantifier ranging everything in the domain. Since for any rigid predicate F $F \subseteq \vee$, it follows that $E_F p \rightarrow E_\vee p$. And since $\Box p$ is just defined as $E_\vee p$, we can derive $E_F p \rightarrow \Box p$, which is (NP) itself. But there is another reason to regard the definition to be unsatisfactory (Romero 2019, p.129-36). As it was mentioned, Fine defines necessary truths as propositions that are true by the essence of all objects. However, by “all objects” he means all objects in the domain, which is the set of all possible objects. Persuasively, since the problem of which objects are possible depends on the notion of metaphysical modality, to conceive necessary truths the essence of all *possible* objects is circular. In the same paper Fine also writes “[...] one might operate with a broader conception of possible object than is customary in modal logic” (Fine 1995a, p.244), but it is still unclear what the “broader conception of possible object” is.

To understand essence in terms of generalized identity may seem to offer better prospects for advocating (NP), since “[a]s with objectual identity, gener-

⁶ $E_\vee p$ may be read: p is an essence of all objects. There are logical machinery behind it in Fine (1995a), but it would be irrelevant here, and I do not want to overcomplicate things.

alized identity is tightly linked to metaphysical necessity” (Correia and Skiles 2017, p.656). Following Correia and Skiles, let “ $p \equiv q$ ” denote the generalized identity between p and q . Here $p \equiv p$ is a theorem⁷ and therefore in any normal modal logic it follows that $\Box(p \equiv p)$. And given $p \equiv (q \wedge r)$, which means that q is a partial essence of p , by substitutability we get $\Box(p \equiv (q \wedge r))$. That is, assuming q is a partial essence of p , it follows that q is a partial essence of p necessarily.⁸

This actually proves something stronger than (NP): namely, $E_F A \rightarrow \Box E_F A$. However, since anything true in virtue of the nature of some object would be true *simpliciter*, this stronger consequence implies (NP). The problem here is that the proof presupposes $\Box(p \equiv p)$: that the generalized identities are necessary. What is the source of the necessity of $(p \equiv p)$? It has been argued that the necessity of $\Box(p \equiv p)$ cannot be explained by the essence of generalized identity (Leech 2021, pp.901-2; cf. Romero 2019, p.138). For example, suppose that the necessity is explained by that *for two things to be generically identical is partially to be in a reflexive relation*. Grant, but what explains the necessity of the partial generalized identity between generalized identity and reflexive relation? Here we cannot appeal to the necessity of $p \equiv p$, for that exactly is the supposed explanandum. Therefore, to appeal to the essence of generalized identity itself does not solve the problem.

Leech seems to conclude that in explaining the necessity of the generalized identity “*one should not appeal to the essence of anything to account for this necessity*”. She further considers this constraint as “the crucial and general objection” on the necessity of essence (p. 902).

I disagree. I think Leech overlooks another possibility: that generalized identities are necessary because of the essence of *to be necessary*. Furthermore, in my view, this also applies directly to (NP). Every truth essential to something or all things are also necessarily true, not because of the nature of essence but because of the nature of what it is to be necessary. In remaining sections, I will argue that the solution for essentialists is to locate (NP) as a partial essence of to be necessary - the thesis I will call (NPEN).

3 The Challenge of Gap Generalized

I argued in section 1 that all essentialists should be non-transmissivists. Typically, in a non-transmissive explanation $\Box p$ because q , the role of q is taken by a

⁷For they take generalized identity to be reflexive (p.645)

⁸It would be clear that in Correia and Skiles’ definition the full essence is trivially also a partial essence.

proposition *about* p . An example would be that p is true in all possible worlds. Let C be any such condition or criterion a proposition can satisfy. A specific non-transmissivist theory on the general source of necessity will then give an account of what the status C is. Thus, non-transmissivists would generally hold that $\Box p$ because $C(p)$, where p necessarily satisfies C . Essentialists are those who hold that the condition C is specified as *to be objectively essential*.

All non-transmissivists should affirm the General Necessity Principle (GNP): $\forall p(C(p) \rightarrow \Box p)$. The reason is same as the reason why essentialists must accept (NP). If there are some true propositions that satisfy C but are not necessarily true, it means that there are something additional to C that makes propositions satisfying C necessary. C is then at most only a part of the condition a proposition must satisfy to be necessary, contrary to the initial assumption.

This indicates that the problem of gap is not peculiar to essentialists. It turns out to be a general difficulty for any non-transmissivist account. For example, a possible world realist would give an explanation: $\Box p$ because p is true in all possible worlds. But if the notion of possible worlds is defined without modal notions, one can equally ask, following Mackie, how possible world realists can “deliver a modal rabbit out of a non-modal hat”. The general difficulty is to provide an elucidation of how a non-modal understanding of C can get along with (GNP) well. The problem now becomes: what can non-transmissivist use for providing the metaphysical basis for (GNP)?

Notice that the statement for (GNP) contains only three elements: the quantifier ranging over proposition or facts; the higher-order predicate C ; and the higher-order predicate $\Box$. It therefore seems to me that if (GNP) is true, it is either essential of *to be necessary*, essential of *to be C* or not essential to any of them.⁹ Below I argue that any suggestions other than the first cannot validate (GNP) by their own.

3.1 Taking (GNP) to be Independently True

First, one *cannot* merely appeal to some fact about C or necessity that is not essential of *to be C* or *to be necessary*, but take $C(p)$ to fully ground $\Box p$. Suppose a proposition satisfying C is necessary as long as C satisfies other non-essential criteria, say X . That is, (GNP) depends only on $X(C)$, but $X(C)$ is not a partial essence of *to be C* nor *to be necessary*. Then, what *explains* $\Box p$ is not merely $C(p)$ but $C(p)$ and $X(C)$, since $C(p)$ cannot guarantee $\Box p$ alone. Additional

⁹I fail to argue against the hypothetical suggestion that (GNP) is essential to the quantifier, for I cannot even comprehend how it would make sense.

component $X(C)$ is required because the nature of to be C or to be necessary does not by itself ensure $X(C)$. So the non-transmissivist account that $\Box p$ because $C(p)$ fails, for to satisfy C turns out not to be everything that is needed to explain necessity.

The argument naturally translates into the case of essentialism. Suppose an essentialist hold that all objectual essences are necessary *only because*, say, any possible world has to obtain every fact that is objectually essential. That is, (NP) would not hold if the fact that any possible world obtains every objectual essences does not obtain. Suppose again that this fact has nothing to do with the nature of *to be objectually essential*. The fact can bear any status as you want. It could be contingent, essential of *to be a possible world*, or even be a metaphysical law - as long as it does not make it essential of *to be objectually essential*.

However, if two suppositions are correct, the necessity of p is not fully addressed by the essentiality of p . If the supposed unique full ground of (NP) does not hold, (NP) would fail; and since (NP) is an extensional statement, its failure implies the existence of contingent objectual essence, and therefore the failure of essentialism. Essentialists should therefore make a further posit that any possible world obtains every objectual essence. And importantly, the further posit should be *explicit*. If it was essential of *to be objectually essential*, then $E_F p$ would by itself guarantee that any possible world would contain p , and perhaps therefore $\Box p$. But our supposition was that it is *not* essential to it.

Hence, we are left with the observation that under the supposition $E_F p$ is *not* the full ground of $\Box p$. If it is not the case that any possible world obtain all objectual essences, there would be some p such that $E_F p$ and $\neg \Box p$. And best, perhaps the only way out, within the spirit of essentialism, is to say $E_F p$ is only a *partial* ground of $\Box p$, where the full ground includes that any possible world obtain all objectual essences. And that is to discard essentialism as it was presented in the first place.

Similarly, we cannot appeal to non-essential property of to be necessary. Suppose (GNP) obtains as long as $Y(\Box)$ obtains and that is neither an essence of to satisfy C nor to be necessary. The correct account then should be that $\Box p$ because $C(p) \wedge Y(\Box)$, and the original account turns out to be untenable.

But what is the problem of adding the auxiliary ground into the account? Besides appearing to be somewhat *ad hoc*, one crucial concern I will note here is that the role of C then becomes inessential to the theory. If $X(C)$ or fully grounds (GNP) and therefore $\Box p$ because $C(p)$ and $X(C)$, it seems what *fundamentally* grounds $\Box p$ is p satisfying *some* condition that satisfies X , where C accidentally becoming the realizer. Similarly, if (GNP) is grounded by $Y(\Box)$,

the essentiality of p becomes the source of p having whatever having Y , not essentially the metaphysical necessity.

One might try to make C or $\Box$ directly relevant by positing yet additional supposition that $E_X X(C)$ or $E_Y Y(\Box)$. From here on the discussion becomes very abstract, and I must confess that I cannot think of any specific candidate of X or Y that plausibly can play all the roles conferred. But still there is an equally abstract worry. Take again $E_X X(C)$ as representative example, where the parallel argument applies for the case of $E_Y Y(\Box)$. It will be argued shortly, that $E_X X(C)$ would grant *to be* X is not metaphysically prior than *to be* C , or that $X(C)$ somehow constitutes what it is to be X . But how a fact about *to be* C - that whatever is C , is necessary - can be grounded on the fact about *to be* X , where *to be* X is constituted in terms of *to be* C ? Of course, since the specific grounding mechanism from $X(C)$ to (GNP) is not specified, it is difficult to formulate a precise argument against the possibility. However, I think this line of thought is sufficient to cast doubt to the suggestion $E_X X(C)$. Who wants to overcome this must (1) specify what the candidate condition for X would be; (2) make clear how such $X(C)$ can ground (GNP); and (3) make clear how, despite the grounding thesis, we can adopt $E_X X(C)$. My claim is that I have trouble even in the early stage of (1) and (2); and (3) seems extremely difficult.

To sum up, if one wants to insist that $C(p)$ provides full explanation for $\Box p$, one cannot think (GNP) obtains regardless of the nature of to satisfy C or to be $\Box$. Notice that the argument does not show it cannot be the case that $X(C)$ or $Y(\Box)$ grounds (GNP). It shows that even if one believes that $X(C)$ or $Y(\Box)$ grounds (GNP), in order to be a non-transmissivist, one should *also* think the essence of to be C or to be necessary is sufficient to guarantee (GNP).

3.2 Taking (GNP) to be Essential of *to be* C

Given the consideration above, it might be tempting to think the case would be solved by positing (GNP) to be essential of *to be* C . However, this alone also would not do. Suppose for the contrary that $E_C(C(p) \rightarrow \Box p)$. Now, if one takes the notion of essence as closely tied to the notion of real definition, it follows that to satisfy C is partially *defined* in a way such that it is the case that $C(p) \rightarrow \Box p$. Or following a revised version of definitional essentialism (Fine 2015), it follows that $\Box p$ is constitutive necessary condition for some p to satisfy C . It seems to me, then, to satisfy C depends partially on to be necessary, and therefore $C(p)$ cannot ground $\Box p$.

This can also be effectively shown in terms of generalized identity. Suppose for the contrary that $C(p) \equiv_p \Box p \wedge q$: to be C just is to be necessary and q .

Now, since a statement of generalized identity closes a theoretical gap (Rayo, 2013); if the left side of a generalized identity statement is the explanans of a metaphysical explanation, the right side must be able to substitute it. A brief example: suppose x 's not being the subject of the marriage loan is metaphysically explained by x 's being a bachelor. And since to be a bachelor just is to be an unmarried adult male, x 's not being the subject of the marriage loan would also be metaphysically explained by x 's being an unmarried adult male. Similarly, given $C(p) \equiv_p \Box p \wedge q$, the claim that $C(p)$ metaphysically explains $\Box p$ is equivalent to the claim that $\Box p \wedge q$ metaphysically explains $\Box p$. And that apparently does not seem satisfactory, therefore the claim that $C(p)$ metaphysically explains $\Box p$ should be discarded.¹⁰

Perhaps the argument can be made clearer by introducing the notion of essential dependence: x depends on y “if y is a constituent of a proposition that is true in virtue of the identity of x ” (Fine 1995b, p.275; the term “essential dependence” follows Correia (2008)). Therefore from $E_C(C(p) \rightarrow \Box p)$, it follows that *to be C* essentially depends on $\Box$, that is, *to be necessary*. Of course essential dependency is not antisymmetric. However, what non-transmissivists need is that *to be C* is strictly prior¹¹ than *to be necessary*, since standard understandings of ground (cf. Schaffer 2009, Fine 2012) take the ground strictly prior, or more fundamental than the grounded.

However, the appeal to essential dependence may seem somewhat awkward. Whereas I am operating with the consequential sense of essence, the notion of essential dependence is usually formulated in terms of constitutive essence, and for good reasons. For example, assume the standard definition of biconditional ‘ $\leftrightarrow$ ’ can be also read as a grounding-essence claim: $\forall p \forall q (((p \rightarrow q) \wedge (q \rightarrow p)) < (p \leftrightarrow q))$ and $E_{\leftrightarrow} \forall p \forall q (((p \rightarrow q) \wedge (q \rightarrow p)) \leftrightarrow (p \leftrightarrow q))$ ¹². The obvious implication would be that the higher-order relations $\rightarrow$ is strictly prior than the higher-order relation $\leftrightarrow$. Yet since $\forall p \forall q (((p \rightarrow q) \wedge (q \rightarrow p)) \leftrightarrow (p \leftrightarrow q))$ is a logical truth, it would be a consequential essence of anything, including $\rightarrow$. And under the supposition it yields that $\rightarrow$ depends on $\leftrightarrow$, contrary to the assumption.¹³

¹⁰A similar argument is anticipated by Romero (2023).

¹¹Following Fine (1995b), I use the term ‘dependency’ for the fundamentality relation that does *not* imply antisymmetry (p.282-3). If X depends on Y , one only can infer that X is *not prior* than Y , for the possibility of Y being dependent on X is still open. The term ‘prior than’ is reserved for the antisymmetric fundamentality relation.

¹²That is, the definition gives *constitutive necessary and sufficient condition* for biconditionals (Fine, 2015). Two separate statements can be unified under the notation presented there: $((p \rightarrow q) \wedge (q \rightarrow p)) \leftrightarrow_{p,q} (p \leftrightarrow q)$

¹³Fine (1995b) tries to solve this problem by introducing the additional condition: “ x depends upon y just in case y cannot be generalized out of the consequentialist essence of x ” (p.278), where y can be ‘generalized out’ of the essence of something if all occurrence of y in the essence can be substituted by any object and still belongs to the essence. This perhaps can solve the problem regarding the dependence between *objects*. Nevertheless, it does not seem to work regarding dependence outside first-order objects. What I proposed seems to be a counterexample:

Now, if (GNP) is proposed as a constitutive essence of *to be C*, the aforementioned worry does not occur - and my argument still works fairly well. But perhaps it is not. For example, non-transmissivists in fact must accept not only the conditional but also the biconditional $C(p) \leftrightarrow \Box p$, for it aims to account the *general* source of necessity. Then given the motivation, one might want to say it is in fact the biconditional which is a constitutive essence of *to be C*, and (GNP) is merely consequential. I find this proposal reasonable. Nevertheless, my argument *still* works fairly well, for ' $\Box$ ' occurs in the biconditional as well.

So we must examine *how* would (GNP) be essential consequentially. The following exhausts the logical possibility: (1) (GNP) is a logical truth; (2) constitutive essence of *to be C* is inconsistent; (3) $\forall p \neg C(p)$ follows from the constitutive essence; (4) $\forall p \Box p$ follows from the constitutive essence; (5) constitutive essence includes $\Box$; (6) constitutive essence includes $\Diamond$.¹⁴

(1) – (4) are untenable, whereas (5) and (6) shows that *to be C* still depends on *to be necessary* or something similar. (1). It would be clear that (GNP) cannot be a logical truth in some logics which is customary - classical first-order logic, normal modal logic, and so on. If (GNP) is a logical truth in that sense, the challenge of gap would not arise at all. Of course, since the main focus here is on essentialism, it should be noticed that (NP) can be proven under Fine's logic of essence supplemented with his definition of necessity as maximal objectual essence. However, first, I see no reason to think the definition is itself a logical truth. Second, we have a reason to withdraw his definition, which was discussed in section 2, and the definition is certainly dispensable from the logic. So even if contrary to the first point, the definition is part of the logic of essence, it would be better to operate with the weaker logic without such a definition. Finally, as I will show in the next subsection, Fine's definition actually yields that (NP) is an essence of *to be necessary*. Hence, the definition is not something who wants to yield that (GNP) is *only*, or *fundamentally* essential of *to be C* can consistently endorse - which is what I am discussing here. Therefore, one cannot appeal to (1) to escape the dependence problem.

(2). Since essence, what is true in virtue of the nature of what is true *simpliciter*, taking (2) means taking the world to be inconsistent. I do not think this is a price many philosophers want to pay, and those who willing to pay the price would typically refuse to say we can infer anything from an inconsistent premise. (3). This means that there are no sources of necessity, and therefore no necessary truths. It is intuitively absurd, and I do not think the task of making sense of the proposal and refuting it would be adequate for the purpose of the

$\leftrightarrow$ in the consequential essence of $\rightarrow$ cannot be generalized out, for substituting into some two-place relation over propositions or facts would not result in logical truths. Also see Vogt (in press)

¹⁴I have classical S5 logic in mind; but it is also exhaustive even if one endorses S4 or intuitionist logic.

paper. At least dialectically, I expect one who enters the debate of the source of necessity would not hold there are no necessary truths. The same consideration applies to (4), which means that every proposition is necessarily true.

I find no independent reason to reject the possibility of (5). However, it would be obvious that if the constitutive essence of *to be C* includes $\Box$, it follows that *to be C* indeed depends upon *to be necessary*. And as I argued, this undermines the original non-transmissivist account. To show (6) results in undesirable consequence is not straightforward. One easy way out, is of course to take $\Diamond$ defined in terms of $\Box$, so that you cannot get (6) without (5). But perhaps one would prefer to take $\Box p$ to be defined as $\neg\Diamond\neg p$, or two operators to be on a par in regard of fundamentality. Throughout the paper, I implicitly assumed that $\Box$ is the fundamental modal operator, and essentialism is an account on the source of *necessity*, not other kinds of metaphysical modality. However, this assumption was made only because it facilitates the discussion. It is certainly possible for any non-transmissivist account to be formulated in terms of possibility: the source of *possibility* of any possible truths q is explained by $\neg C(\neg q)$. If $\Diamond$ is more fundamental than $\Box$, non-transmissivism probably should undergo such a modification. But once the modification is made, (6) undermines the account, for it implies that *to be C* is dependent upon *to be possible*.

Hence, utilizing the essence of *to be C* to vindicate (GNP) would make it essentially dependent on *to be necessary*. Still it might be argued that that does not preclude *to be C* grounding *to be necessary*. The preclusion is by the following reasoning. (i) If *to be F* essentially depends on *to be G*, *to be F* is not prior than *to be G*; (ii) If *to be F* grounds *to be G*, *to be F* is prior than *to be G*. Conclusion: if *to be F* is a constituent of the essence of *to be G*, *to be F* cannot ground *to be G*. The criticism is that for the inference to be valid, “prior than” in (i) and (ii) should express the same metaphysical relation, which does not seem likely (Vogt, 2026).

But to draw the conclusion one need not hold that (i) and (ii) operates with the identical notion of priority. To show that the priority cannot head in the opposite direction would suffice. This would be done, for example, if we show that both grounding and essence gives constitutive condition, where presumably constitution cannot be circular (Fine, 2015). Also, one can argue that the real definition - which is sometimes taken to be the constitutive and immediate essence - of whatever can be formulated in terms of grounding (Rosen 2015, pp.198-9¹⁵). I am inclined to think the account of Fine (2015) is largely correct, but to argue for the macroscopic metametaphysical theory would certainly

¹⁵Notice that the final proposal in p.200 is strikingly similar with Fine (2015): a real definition giving a necessary and sufficient condition must involve both essence and ground.

exceed the scope of the paper. Still, some direct arguments for the opposite-heading of essence and ground should be dealt.

Vogt (2026) suggests that, assuming Aristotelian account of property, there can be cases where Pa grounds *property P exists*, whereas E_aPa (p.519). Vogt seems to regard this being the case that a being both prior (by grounding) and not prior (by essential dependence) than P . However, it should be noticed that ' P ' is used once as a name for a first-order object, once as a predicate. Thus, what the grounding statement and the essence statement jointly imply is actually that a - or more rigorously *to be identical to a* - is not prior (by essential dependence) than *to be P* ; where a is prior (by grounding) to the *property P* . This is clearly not a case of two priority relations heading the opposite direction.

Vogt (2026, in press) also cites various specific cases where one might contend that Fx grounds Gx but $E_F(Fx \rightarrow Gx)$ or $E_{\lambda p.p=Fx}(Fx \rightarrow Gx)$ (see also Correia (2018)). Of course, she is cautious enough not to declare any of the particular example to be definitive. However, it is suggested that the candidate examples, taken together, has enough force to make one to take such possibility seriously. I disagree, for it seems that, even though that the grounding claim and the essentiality of the conditional is correct, those putative examples are subject to a uniform treatment. That is: the conditional should be the essence of the grounded, not the ground. For example, Vogt holds in both paper that *a promises that p grounds a has the obligation to p* , whereas it seems that it is essential of *to promise* that if one promises that p then one has the obligation to p . One of course may deny that there is such an intimate connection between *to promise* and *to have obligation*. If there is, my treatment is to deny the essence claim, and alternatively say it is essential of *to have obligation* that if one promises that p then one has the obligation to p .¹⁶¹⁷

Therefore, insofar as non-transmissivists tries to *ground to be necessary* - alternatively, *to be possible* - in their preferred condition *to be C* , they should not think (GNP) can be metaphysically vindicated solely by the nature of *to be C* . Essentialism, *a fortiori*, should not think (NP) is vindicated solely by the nature of *to be objectively essential*. In fact, what I rejected is what Wilsch (2017) and Wallner and Vaidya (2020) already suggested. Wilsch directly takes

¹⁶This is exactly parallel to the goal of this section. Non-transmissivists would agree that $C(p)$ grounds $\Box p$. But if one also have to think that the conditional $C(p) \rightarrow \Box p$ should be essential, then it should be essential of the grounded, that is, *to be necessary*, not of the ground, that is, *to be C* .

¹⁷One might criticize such a treatment to be *ad hoc*. It perhaps is, but notice this does not increase the theoretical complexity of priority, but reduces it. If the priority by grounding and priority by essential dependence was so fundamentally different so that it permits opposite-heading, it seems that there should be two fundamentally different account of what metaphysical priority is, perhaps even two different kinds of fundamental reality, which should be altogether compatible. That not being present, I would regard the *ad hoc* move justified.

(NP) to be the essence of essence, whereas Wallner and Vaidya takes both $E_x p \rightarrow \Box E_x p$ and $E_x p$ can explain $\Box p$ essential to the essentialist operator E_x . Since $E_x p \rightarrow \Box E_x p$ entails (NP), it would be fair to say both takes (NP) as a mediate and consequential essence - which I simply say ‘essence’ - of *to be objectually essential*.

In case of Wilsch, he explicitly states himself as a modal primitivist, not an essentialist. Although he takes essence to give the “source” of metaphysical necessity, here what the “sources” actually do is to determine the extension of various kinds of necessity, and account for the naturalness of them. “The sources of necessity are the modal joints of nature.” (Wilsch 2017, p.429). Essence, then, explains why metaphysical necessity is a significant kind of primitive necessity, why it “carves the nature at its joints”. That is apparently a different enterprise from the problem of source Hale has in mind, and what is being discussed in this paper.¹⁸

Yet Wallner and Vaidya take themselves as *non-reductive Finean essentialists*, “which endorse [...] that modality is grounded in definitional essences.” (Wallner and Vaidya 2020, p. 431). They also make it explicit that they intend to follow Hale in giving non-transmissive explanation for necessity. Then my argument above shows that it makes *to be objectually essential* no prior than *to be necessary*. And that seems to be in tension with the claim that modality is grounded in essence. Also, if one is sympathetic with Correia and Skiles, one can further criticize the account for offering a trivial explanation on necessity, in the form $\Box p$ because $\Box p$ and q .

3.3 Taking (GNP) to be Essential of *to be Necessary*

The only remaining possibility for non-transmissivists, then, is to say that the relevant instances (GNP) is an essence of *to be necessary*: $E_{\Box} \forall p (C(p) \rightarrow \Box p)$. By doing so one can insist that *to be C* suffices for the full source of necessity. For in this case nothing else than the nature of *to be necessary* is required to guarantee that (GNP) holds, and therefore one need not posit a further source of necessity that validates the leap from $C(p)$ to $\Box p$. It also does not make *to be C* not prior than *to be necessary* and thus making the grounding claim suspicious. By placing (GNP) into the essence of *to be necessary*, it follows that *to be necessary* is not prior than the constituent of its essence: *to be C*,

¹⁸My argument above does show that Wilsch cannot be a non-transmissivist. But that is somewhat trivial, for a modal primitivist would deny there is a “source” of necessity in our sense. Wilsch’s account is best conceived as a complete rival for essentialism - not a variant of it - just as deflationism or dispositionalism is. And to advocate essentialism among a variety of theories would exceed the scope of this paper.

which is in harmony with the grounding claim that $C(p)$ grounds $\Box p$.

Since this is the only option left for non-transmissivists, they should take the relevant instance of the essence claim into the account. For example, possible world realists must say it is somehow the nature of *to be necessary* which guarantees that whatever true in all possible worlds are necessary. Applied to essentialism, the claim would be that it is by the essence of *to be necessary* that whatever is objectively necessary is necessary. Below I will abbreviate this claim as (NPEN)¹⁹ and the generalized version (GNPEN). That is:

$$(NPEN) \quad E_{\Box} \forall F \forall p (E_F p \rightarrow \Box p)$$

$$(GNPEN) \quad E_{\Box} \forall p (C(p) \rightarrow \Box p)$$

where again F rigid.

And in fact Fine's original account turns out to imply (NPEN), although not explicitly stated. As I mentioned earlier, Fine *defines* $\Box$ in terms of essence: $E_{\Box} p \leftrightarrow_p \Box p$. And its implication $E_{\Box} p \leftarrow_p \Box p$ is just another, intuitive way to put $E_{\Box} \forall p (E_{\Box} p \rightarrow \Box p)$ (Fine 2015, p.299). I also mentioned that for any rigid predicate F $E_F p \rightarrow \Box_{\Box} p$. Since from $E_{\Box} p \rightarrow \Box p$ and $E_F p \rightarrow \Box_{\Box} p$ it follows that $E_F p \rightarrow \Box p$; we can get $E_{\Box} \forall F \forall p (E_F p \rightarrow \Box p)$.

But I also mentioned that Finean definition is accused to be circular, for he presupposes the domain of *all metaphysically possible* objects. An additional concern is that we might want to talk about the essence of *impossible* objects, which are not in any broad sense of "possible object" one of them. Typical examples is the Russellian set. Why is the Russellian set impossible? An essentialist might want to argue as follows. The Russellian set is essentially a set of all sets that does not contain itself; therefore it is necessary that for every x , if x is identical to the Russellian set, it is a set of all sets that does not contain itself. And it is also necessary that, there is no x such that it is the set of all sets that does not contain itself. Therefore, Russellian set necessarily does not exist, i.e., is an impossible object. But for such an argument to be sound, it should be held that the essence of an impossible object can also be the source of some necessities.

But then how could the Finean definition be maintained? One way out would be to construct a domain which includes both possible and impossible objects. But which objects, possible or impossible, are there? Perhaps Meinongians could give us an answer.²⁰ However, I think it is an unnecessary detour to think

¹⁹Necessity Principle as the Essence of to be Necessary

²⁰See, for example Zalta (1983) or Priest (2005). Fine himself also worked on the problem of non-existent and

that essentialists would need a theory of objects.²¹ It is not very desirable for essentialists to just wait until object theorists come up with a comprehension principle of objects everyone can happily endorse.

Instead, I suggest discarding the Finean definition, and take (NPEN) not as a consequence of the definition, but directly into the essentialist account on necessity. Finean intuition would then be preserved under any suitable theory of objects, as long as the axiom $F \subseteq G \rightarrow (\Box_F p \rightarrow \Box_G p)$ is employed.

But not only Fine, but all essentialists must accept (NPEN). Otherwise, either essence turns out not to provide a full explanation for necessity; or essence cannot ground necessity because it infringes priority or irreflexive constraint of ground. One consequence is that any compelling argument for essentialism would also be an argument for (NPEN). For I do not aim in this paper to argue for the essentialist answer against other alternatives, I will not explore possible specific arguments for (NPEN).

What I *should* show in this paper, is how essentialists can answer the challenge of gap by utilizing (NPEN). To briefly reiterate, their main criticism was that once we stick to the notion of essence that is *not* defined in terms of necessity, there seems to be a gap between essence and necessity. My suggestion is, of course: the gap is to be bridged by the essence of to be necessary.

Let us return to Hale's argument for the necessity of definitional essence. I take Romero's criticism to be correct that we cannot infer $\Box E_\alpha \Phi \alpha$ from $E_\alpha \Phi \alpha$ without appealing to the essence of *to be necessary*. To say $E_F p$ where F rigid is certainly not to say $\Box E_F p$. The gap, however, can be filled by the nature of what it is to be necessary, and an independent argument for the necessity of objectual essentiality can be given. By (NPEN), *T-axiom*²² and modus ponens we get $\forall F \forall p (E_F p \rightarrow \Box p)$ where F rigid. By universal instantiation we get $E_F E_F p \rightarrow \Box E_F p$, and together with the theorem $E_F p \rightarrow E_F E_F p$ where F rigid, we can cut out the middleman and infer $E_F p \rightarrow \Box E_F p$, where F rigid.

Even better, we can revise Hale's argument in which does not depend on $E_F p \rightarrow \Box E_F p$. Suppose again, that $E_\alpha \Phi \alpha$. Now, Romero and Leech ask: why β such that $\neg \Box \Phi \beta$ cannot be identical to α , if the initial presupposition was only that it is *actually* $E_\alpha \Phi \alpha$? Now we have a further reply for that. Suppose for contradiction that $\alpha = \beta$ and therefore by substitutability $\neg \Box \Phi \alpha$.

the comprehension principle on objects. The negative part, which he criticizes certain kinds of object theories, was published in Fine (1982). There he also stated that the positive project where he develops his own theory will be published as the part II of the paper, but unfortunately it remains as a non-existent object till now.

²¹It is even unclear if an independently developed object theory would get along with essentialism well. For example, Zalta, although I see no logical inconsistency between his theory of objects and essentialism, advocates a modal conception of essence specified in terms of his theory of objects in Zalta (2006).

²² $E_F p \rightarrow p$ (Fine 1995a, pp.247-8)

Now, from (NPEN) we can prove, utilizing the *T-axiom* and modus ponens that $\forall F \forall A (E_F A \rightarrow \Box A)$. By universal instantiation we get $E_\alpha \Phi \alpha \rightarrow \Box \Phi \alpha$; and by the assumption $E_\alpha \Phi \alpha$ and modus ponens we get $\Box \Phi \alpha$. Contradiction. Therefore, there can be no β such that $\alpha = \beta \wedge \neg \Box \Phi \beta$, which ensures that if Φ is actually an essential property of α , then it is a metaphysically necessary condition for something to be identical to α .

Notice that this is nevertheless not to ignore the essentialist intuition that it is in some sense “obvious” that objectual essences are necessary. I suggested that essentialists cannot take objectual essence to be *ab initio* necessary. However, it seems also true that there certainly is some kind of frustration when essentialists are confronted with questions like “why are facts of objectual essential necessary?” or “why an object necessarily has the property that defines what the object is?”

This is, I suggest, same kind of frustration when one is confronted with questions like “why is water H_2O ?” or “why does Socrates’ singleton set have Socrates as member?”. As Glazier (2017) argues, we can simply answer: “that is just what water is” or “that is just what the set is”. Turn to the question “Why are objectual essences necessary?”, I suggest one kind of answer essentialists can give is: “that is just what to be necessary is!”. And to give such an answer is to accept (NPEN).

So much for the definitional conception of essence. If essence is understood in terms of generalized identity, we can translate the same idea into $\exists q (\Box p \equiv_p (\forall x \exists r ([\lambda y. y = x] z \equiv_z (p \wedge r))) \wedge q)$. In short: to be necessary *just is* to be a partial essence of some object. We can now proceed similarly as we did in the case of definitional essence. How can we argue here that therefore all objectually essential facts are necessary? Suppose we have a statement of objectual essence: $[\lambda y. y = a] x \equiv_x (s \wedge w)$. s is a partial essence of the object a . And it follows from the nature of to be necessary that for s to be necessary is just to be a partial essence of a . Since s is a partial essence of a , we get that s is necessary.

Interestingly, here we have another option. Recall that in section 2, the charge was not on (NP), but on $\forall p \Box (p \equiv p)$. Can we account for this stronger thesis than (NP) instead? I think yes. Since we cannot say $\Box (p \equiv p)$ is true by the essence of the generalized identity, the core idea again is that it has something to do with the essence of to be necessary.

But we still should be careful with the formulation. For example, $\exists q (\Box p \equiv_p \forall r (\Box (r \equiv r) \wedge q))$ might seem natural; but it would not work. For one thing, the supposition is consistent with there being no necessary truths yet there being true instances of generalized identity. It is also quite mysterious why should the

nature of a particular fact having necessity involve that any fact of generalized identity is necessary.

Instead, I suggest: $\exists q(([\lambda G.Gp \equiv_p \Box p]F) \equiv_F (\forall p \Box(p \equiv p) \wedge q))$. This, however, is not to say it is the generic essence of *to be necessary* that $\Box(p \equiv p)$. It states that it is the essence of *to be just is to be necessary* that $\Box(p \equiv p)$ ²³. In this case, it is not involved in the essence of some particular proposition to be necessary that any fact of generalized identity is necessary. $\Box(p \equiv p)$ is also guaranteed, for $\Box p \equiv_p \Box p$ is a theorem. Now essentialists can proceed their argument as it was suggested in section 2. We can by substitutability get $\Box(p \equiv (q \wedge r))$ and $\Box(p \equiv q)$, which means that the fact on what is essential to what is necessary.

4 Circularity

I argued in the previous section that the only viable option for non-transmissivists is to take the relevant instances of (GNP) to be a partial essence of to be necessary. This was done by showing that any other alternative is incompatible with non-transmissivism. But to say that it is the only viable option for non-transmissivists is not to say that it is tenable. After all, non-transmissivism might also be incompatible with (GNPEN); and with the argument I presented it completes a trilemma which likely undermines any non-transmissivist theory.

It seems clear that (GNPEN) is not *logically* inconsistent with non-transmissivism. To say what it is to be necessary is such and such is one thing, and to say the necessity is explained by such and such factor is another. Similar to essentialism, which is an instance of non-transmissivism.

But it could be the case that the thesis seriously undermines the theoretical motivation of non-transmissivism. The reason of rejecting (GNP) being essential of *to be C* was not logical inconsistency. The argument against it was that it gives a verdict on priority which is inconsistent with non-transmissivism.

As I noted earlier, (NPEN) would not face the same problem. But perhaps there are other problems. Romero (2019) states six constraints on the notion of explanation which the ‘explanation’ of modality by essence must satisfy:

Why The explanation must answer to a ‘why’ question.

Acyclicity The explanation should not be circular.

²³I suspend to judge whether to regard predicates involving higher-order identities, such as $\lambda G.Gp \equiv_p \Box p$ as rigid. I prefer not.

Fundamentality The explanans should be metaphysically more fundamental than the explanandum.

Ontic and Factive It must be the case that the truth of the explanans, not our concept or theory of it, explains the truth of the explanandum.

Structural The explanandum need not be an object or a property.

Objectivity The explanation should be objectively valid. (p.124)

It is already shown that (NPEN) does not make essentialism infringe **Fundamentality**. Essentialism with (NPEN) also easily satisfies **Why, Ontic and Factive**. It answers the question “*Why is p necessary?*”, by appealing to the *facts* about essence of things, not our concept of essence. Essentialists would certainly hold that if the essentialist account on necessity is valid, then it is **objectively** valid. Finally, **Structural** is not a negative constraint that one can infringe.

However, whether essentialism with (NPEN) is compatible with **Acyclicity** is perhaps not so obvious. Some might feel uncomfortable to the idea that the nature of *to be necessary* validates the explanation from essence to necessity. But if it is indeed circular, what exact part of the picture contains a circle?

Notice that I do not try to explain the necessity of p by (NPEN). The explanatory work is done by the fact that p is objectually essential. However, it is even unclear why involving (NPEN) in particular explanations would be viciously circular. Suppose I am wrong, and every explanation on necessity must include (NPEN). p is necessary because p is essential to some objects *and every such truths are necessary in virtue of the nature of to be necessary*. What is wrong with this kind of explanation? My point can perhaps be made by a simple analogy. If someone asks me why 3 is a prime number, I can simply answer “because 3 is a natural number that is not a product of two natural numbers smaller than itself.” However, if someone thinks this is incomplete, then I can add: “... *and every such natural numbers are prime in virtue of the nature of to be prime*”. It would be obvious that neither explanation for the fact that 3 is prime is viciously circular.

Perhaps another kind of charge can be made by asking how can one insist that (NP) is necessary. It might seem plausible to think that (NP) is also necessarily true, or at least essentialists must so insist. And if (NP) is necessary, by essentialism there would be some *object n* such that $E_{\lambda x.x=n}(\text{NP})$, hence $E_{\lambda x.x=n}(\text{NP})$ grounds $\Box(\text{NP})$. Persuasively the object n would be something relevant to necessity, and a further posit that the nature of n is grounded by the nature of *to*

be necessary does not seem farfetched. But then $\Box(\text{NP})$ would be grounded by (NPEN), that is, $E_{\Box}(\text{NP})$. And this would seem somewhat problematic.²⁴

Firstly, it needs to be clarified whether essentialists are really committed to the necessity of (NP). I have formulated the account in a way that avoids such a commitment. Essentialists must hold that every fact essential to some *objects* is necessary. However, my proposal was that (NP) is the *generic* essence of *to be necessary*, not an objectual essence of the concept of necessity. Thus, (NPEN) does not force essentialists to admit that (NP) is necessarily true. S4 principle: $\Box p \rightarrow \Box\Box p$ can also be derived without assuming (NP) is necessary, as long as one is willing to accept that (NP) is *logically* true in the logic of essence. And I showed in §3.3 that (NP) logically follows in Fine (1995a), although it is better to implement (NP) directly into the logic.²⁵

Nevertheless, suppose we somehow grant that (NP) is necessarily true, and $\Box(\text{NP})$ is grounded by (NPEN). Still, I must confess that I do not see anything problematic here. To use another analogy: suppose a quasi-God (*‘Q’* below) knows all and only essential facts, and its knowing a essential fact is metaphysically grounded by that fact being essential. And suppose it is essential of *to know* that if someone knows that *p*, it is the case that *p*, that is, $E_K\forall s\forall p(K(s, p) \rightarrow p)$. It would then follow that $E_K\forall p(K(Q, p) \rightarrow p)$. Now, by the assumption on *Q* it would be the case that $K(Q, \forall p(K(Q, p) \rightarrow p))$, and it is partially grounded by $E_K\forall p(K(Q, p) \rightarrow p)$. It seems clear there is nothing viciously circular here. But notice that the proposed situation of $E_{\Box}(\text{NP})$ grounding $\Box(\text{NP})$ is structurally similar to the case of the quasi-God: there being a Φ containing *X* such that $E_X\Phi$, where $E_X\Phi$ grounds $X(\Phi)$. This does open the possibility that in some cases, *to be essential of to be X* can be prior than *to be X*. But this is not an obvious problem, for still the grounding is irreflexive. And if, through metametaphysical investigation philosophers somehow found out that *to be essential of to be X* cannot be prior than *to be X*, essentialists can simply insist that (NP) is not necessary.

²⁴I believe this is what Jonathan Hricko meant when he suggested to me that if *to be necessary* has an essence, then necessity cannot be reduced to what is true in virtue of all essences. If the word is taken literally, the obvious reply is that I do not reduce necessity to what is essential, but to what is *objectually* essential.

²⁵Proof for S4. Suppose $\Box p$, and by essentialism that for some rigid predicate *F* $E_F p$. We infer $E_F E_F p$ as usual. Given (NP) is logically true, its instance would also be logically true, that is, $\vdash E_F p \rightarrow \Box p$. THEOREM 4. (ii): if $\vdash A \rightarrow B$ then $\vdash |B| - |A| \subseteq cF \rightarrow (E_F A \rightarrow E_F B)$ is proven (Fine 1995a, p.253). In ordinary terms, $|B| - |A|$ is the rigid predicate that is satisfied by exactly all those objects occurring in *B* which does not occur in *A*. It is irrelevant which predicate *cF* is, for obviously there are no new objects in $\Box p$ other than objects already occurring in $E_F p$, and thus $\Pi x(|\Box p| - |E_F p|)x$ is always false and therefore $\Pi x((|\Box p| - |E_F p|)x \rightarrow cFx)$ is always vacuously true, regardless of what *cF* is. (formally, $cF =_{df} \lambda x(x \leq F)$ where $x \leq F =_{df} \Sigma y(Fy \wedge x \leq y)$, Σ the existential quantifier ranging all possible objects and $x \leq y$ for ‘*y* essentially depends on *x*’) Together with $\vdash E_F p \rightarrow \Box p$ and the obvious fact that $\vdash |\Box p| - |E_F p| \subseteq cFx$, by THEOREM 4. (ii) we can derive $E_F E_F p \rightarrow E_F \Box p$. Since we had $E_F E_F p$, by modus ponens we get $E_F \Box p$, and by (NP) we get $E_F \Box p \rightarrow \Box\Box p$. Again by modus ponens we get $\Box\Box p$. Therefore, assuming $\Box p$, we can derive $\Box\Box p$, hence $\Box p \rightarrow \Box\Box p$.

5 Conclusion

I argued that essentialists, facing the problem of gap, must respond by accepting (NPEN) - that (NP) is true in virtue of the nature of to be necessary. This follows from the more general character of non-transmissivism. Not only essentialist theories on the source of necessity, but all non-transmissivist theories face the problem of the gap. When one asserts that p is necessary because $C(p)$, one has to elucidate the notion of C without modal notions, and at the same time grant (GNP): $\forall p(C(p) \rightarrow \Box p)$. I argued in section 3 that all non-transmissivists should regard (GNP) to be a partial essence of to be necessary. And since all essentialists theories are non-transmissivist - this is argued in section 1 - essentialists are forced to accept (NPEN). I also tried to show that this account is not circular as it may first appear.

What I did *not* argue in this paper is that essentialism is superior to its competitors. My aim in this paper has been to show that the particular challenge to essentialists can be met coherently without dismissing the challenge. Although what the challenge from gap points out is correct, it is not strong enough to defeat essentialism. It rather helps essentialists to clarify what comes together with essentialism. Therefore, one can read this conclusion as a conditional: *if* essentialism is correct, then so is (NPEN). Any argument in favor of essentialism is also an argument for (NPEN). Furthermore, if one agrees that *essence precedes existence* (Lowe, 2008), it illuminates another argumentative strategy for essentialists. A certain kind of necessity, existent or non-existent, is specified by (NPEN). The remaining task for essentialists would be to argue *that* necessity is the metaphysical necessity. But of course, the contrapositive of the conclusion also holds. If (NPEN) is false, so is essentialism. Any criticism against (NPEN) would thereby be a criticism for essentialism. However, fleshing out new arguments for essentialism, or dealing with those criticisms would exceed the scope of this paper.

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